

f/ Aperture

Synthetic data generation and validation for enterprise AI teams

Generate

Validate

Trust

THE PROBLEM

Real data is the bottleneck for enterprise AI.

ML and AI teams in healthcare, finance, and consulting need large, high-quality datasets to fine-tune and evaluate models. But the real data is either too sensitive to share, too small to be useful, or too expensive to label at scale.

Sparse

Rare clinical cohorts, fraud cases, edge scenarios are underrepresented.

Sensitive

PII and regulated content cannot leave the institution unmodified.

Slow

Data-access waiting periods kill momentum on prototypes and demos.

FROM USER INTERVIEWS

“It is unclear where the data is coming from and why the user can trust this.”

“How do you test that it is good synthetic data?”

“What happens if it is wrong or misleading?”

Five reviewers flagged validation, not generation, as the bigger pain.

Generate validated synthetic datasets — without exposing raw records.

Aperture turns limited, sensitive, or hard-to-access data into validated synthetic datasets for fine-tuning, evaluation, stress-testing, and prototyping.



01

Describe

Plain-English prompt or upload a small real sample.



02

Generate

LLM schema inference + SDV correlation-preserving synthesis.



03

Validate

Fidelity, utility, and privacy checks before export.



04

Export

CSV / JSONL / Parquet — slots into the existing ML pipeline.

Two generation paths, both production-ready.



PATH A

Prompt-only

AI-inferred schema · no real data

Prompt-to-Action Agent (Claude under the hood) interprets the request, infers a schema, and asks clarifying questions when underspecified.

Each column is synthesized independently against an inferred distribution — fast, controllable, but no inter-column relationships.



PATH B

Upload-grounded

SDV GaussianCopula · correlation-preserving

User uploads a small real CSV. Aperture profiles it, fits a copula on the joint distribution, then samples new rows.

Generates a 72-year-old patient with a higher HbA1c because the real data showed it — the synthesizer learned the relationship.

Validation pass on every generation — three axes today.

1

REALISM

Mean / std drift on numerics, category coverage on enums. Averaged across columns.

2

DIVERSITY

Coefficient of variation match. Catches the classic variance-collapse failure mode.

3

SAFETY

Regex scan for email, phone, SSN patterns. Binary — 100 if clean, 65 if any hit.



HONEST CAVEAT

This is marginal fidelity only. There is no measure of whether the data actually helps a model train, and no real privacy check beyond regex.

Serious frameworks validate on three axes. We cover a slice of one.

Industry consensus across AWS, SDMetrics, and clinical-validation literature:



Fidelity

Does it look like the real data?

PARTIAL

Per-column drift and category coverage shipped.
Inter-column correlation, mutual information,
and feature-importance order all missing.



Utility

Does training on it help models?

MISSING

TSTR (train synthetic, test real) and TR+STR
(augmented training) are the gold standard.
Aperture exposes neither today.



Privacy

Can someone reconstruct real records?

MINIMAL

Regex scan catches accidentally-leaked PII. Exact
match, distance-to-closest-record, and
membership inference all missing.

Beyond GaussianCopula — toward task-aware, multi-table, mixed-modality synthesis.

Today we ship SDV's GaussianCopulaSynthesizer. It is fast and explainable, but it has a ceiling.

LIMITS OF THE CURRENT APPROACH

-  **Tail collapse**
Copulas underrepresent the rare events (p95+) that matter most for fraud, severity, and rare clinical cases.
-  **Single-table only**
Real ML data is relational — claims join to policies join to customers. Joint distribution per table is not enough.
-  **Numeric-friendly**
Free-text columns (claim notes, adjuster summaries, clinical narratives) are out of scope.

WHAT WE BUILD NEXT

-  **Multi-table relational synthesis**
SDV HMASynthesizer — preserve foreign keys and cardinality across claims / policies / customers.
-  **Tail-aware sampling**
Conditional sampling that explicitly oversamples the rare class or right tail, with the source ratio recorded for honesty.
-  **LLM-assisted text columns**
Pair structured row with a generated claim_note or clinical_summary — grounded in the structured fields to prevent hallucinated facts.
-  **Task-conditioned generation**
User picks a target column at generation time; the synthesizer optimizes for downstream utility, not just fidelity.

Fidelity: replace ad-hoc drift with citable statistical tests.

CHECK	APERTURE TODAY	INDUSTRY STANDARD	EFFORT
Per-column drift	Ad-hoc mean/std formula	Kolmogorov–Smirnov test (scipy.stats)	1 day
Category coverage	Implemented	Histogram similarity	1 day
Inter-column correlation	Missing	Mutual information score	2 days
Top-N feature importance	Missing	Compare on real vs synthetic	3 days
Tail preservation	Missing	p90 / p95 / p99 deltas (performance diffs)	2 days

AWS specifically recommends mutual information over Pearson correlation — Pearson misses non-linear dependencies (zip code → fraud rate is not linear). SDMetrics ships CorrelationSimilarity as a one-line import.

Utility: the metric the ML engineer actually wants. Missing today.

“Will this data actually improve my model?” — the question fidelity scores cannot answer.

THREE TRAINING CONFIGURATIONS, ONE REAL TEST SET

TRTR

Real → real

Train on real, test on real. The baseline ceiling — what a model trained without synthetic data achieves.

TSTR

Synthetic → real

Train on synthetic only, test on real. How close to baseline = how usable synthetic alone is.

TR+STR

Augmented → real

Train on real + synthetic combined, test on real. The configuration enterprise users actually want.

THE METRIC THAT MATTERS

Rare-class recall lift = $\text{recall}(\text{TR+STR}) - \text{recall}(\text{TRTR})$. For insurance fraud, healthcare rare cohorts, and minority-class problems, this number tells the buyer in one figure whether Aperture earned its keep.

Privacy: regex is the weakest check. Enterprise asks for more.

THE FAILURE MODE REGEX MISSES

A 72-year-old female in ZIP 94305 with type 1 diabetes and an unusual lab panel...

...is plausibly one real person. The synthesizer may have memorized them. Regex sees no email, phone, or SSN — so it scores 100. That is a false sense of safety.

WHAT ENTERPRISE PROCUREMENT ASKS FOR

k-anonymity, distance-to-closest-record histograms, and membership-inference resistance — not regex.



Exact match rate

Set intersection between synthetic and source. Must always be zero — surfaced as a hard gate, not a score.



Distance to Closest Record

For each synthetic row, nearest-neighbor distance to real rows. Histogram, with rows below threshold flagged for review.



Membership inference

Can an attacker tell whether a specific record was in the training data? Run a shadow-model attack as a stress test.

Horizontal validation is parity. Vertical business rules are the differentiator.

Generic synthesizers (Gretel, MOSTLY AI) ship the statistical metrics. They cannot tell you a deductible exceeds its coverage, or that a male patient has a pregnancy code. The rules a generic tool will never know are the moat.



INSURANCE PACK

- › `deductible ≤ coverage_amount`
- › `claim_amount ≥ 0`
- › `if claim_count == 0 then claim_amount == 0`
- › `premium` increases monotonically with `risk_score`
- › `policy_end_date > policy_start_date`
- › `fraud_rate` within source $\pm 0.5\%$



CLINICAL PACK

- › `HbA1c > 6.5` implies diabetic diagnosis
- › `metformin` requires T2D or pre-diabetes
- › no pregnancy code on male sex
- › medication-condition consistency check
- › ICD-10 + procedure code pairing
- › pediatric dosing thresholds enforced

Six use cases. Fraud detection is the wedge.



LEAD DEMO

Insurance fraud detection

Rare positive class (~1.8%), clean augmentation story. +17 pt recall lift on real test set is the target headline.



Claim severity prediction

Tail-aware generation. Report p90 / p95 / p99 separately, not just mean. Warning: 'p99 understated by 23%'.



Healthcare rare cohorts

HbA1c > 12 with comorbidities. Clinical rule validation. HIPAA quasi-identifier uniqueness check.



Customer churn

Sensitive behavior data, rare-combination cohorts (premium hike + competitor quote + late payment).



Finance stress testing

Synthetic data as evaluation, not training. Adversarial scenarios for internal compliance agents.



Consulting prototypes

Bridges the 2–6 week dead zone between SOW and data access. Trust report is the deliverable.

THE BET

The defensible product is the trust layer, not the generator.

SDV ships a copula in one line. Anyone can wire that up. What no one has shipped is a single workflow that generates, validates against utility and privacy gold standards, applies vertical rules, and produces a report a regulated buyer will actually sign off on.

+17pt

Rare-class recall lift on the fraud demo.

100%

Business-rule compliance on insurance and clinical packs.

0

Exact-match privacy violations across every export.